MATH 152
Mrs. Bonny Tighe

# QUIZ 1

6.1
25 points

NAME Solutions

TA_____ Wed 9/5/2018

1. Find the area of the region enclosed by the given curves. Sketch the graphs.

a) $y = \tan 2x,\ y = 1$ *and* $x = 0$

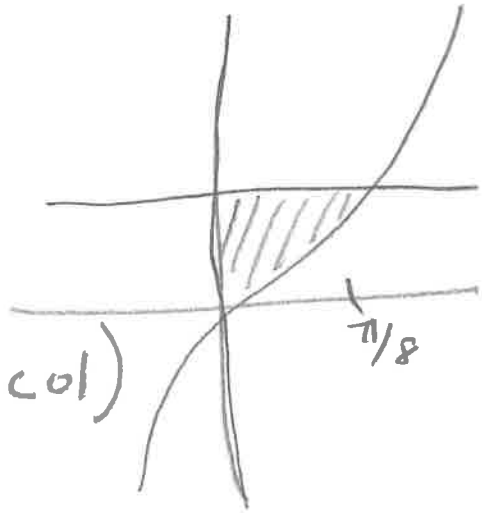
$\tan 2x = 1$
$2x = \pi/4$
$x = \pi/8$

$$\int_0^{\pi/8} (1 - \tan 2x)\,dx$$

$$x - \tfrac{1}{2}\ln|\sec 2x| \Big]_0^{\pi/8}$$

$$\left(\pi/8 - \tfrac{1}{2}\ln|\sec \pi/4|\right) - \left(0 - \tfrac{1}{2}\ln|\sec 0|\right)$$

$$\pi/8 - \tfrac{1}{2}\ln|\sqrt{2}|$$




b) The region bounded by $x = y^2 - 6y$ *and* $x = 4y - y^2$

$y^2 - 6y = 4y - y^2$
$2y^2 - 10y = 0$
$2y(y-5) = 0$
$y = 0 \quad y = 5$

$$\int_0^5 (4y - y^2) - (y^2 - 6y)\,dy$$

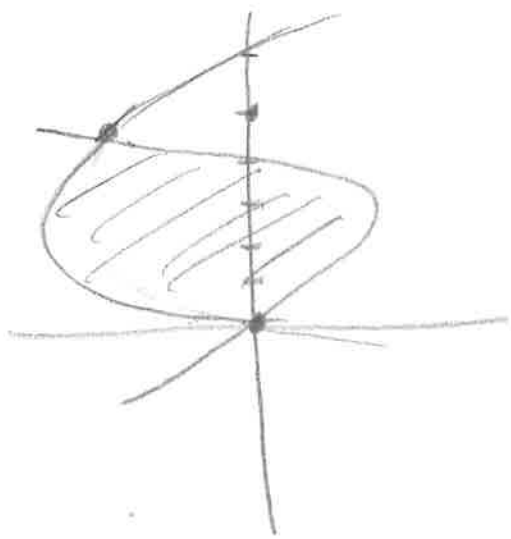
$$\int_0^5 (10y - 2y^2)\,dy$$

$$5y^2 - \tfrac{2}{3}y^3 \Big]_0^5$$

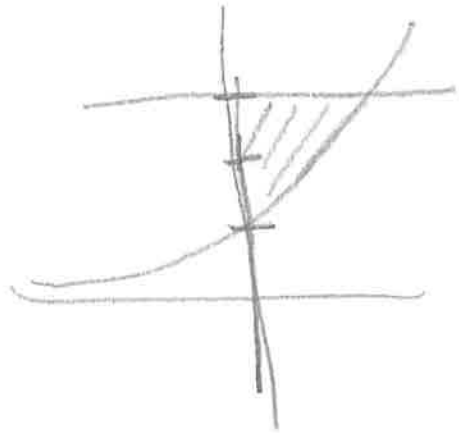
$$5(25) - \tfrac{2}{3}(125) - 0$$

$$125(1 - 2/3)$$

$$125\left(\tfrac{1}{3}\right)$$



c) $y = e^{3x}$, $y = 3$ and $x = 0$ Sketch the graph



$$\int_0^{\frac{1}{3}\ln 3} (3 - e^{3x})\,dx = 3x - \frac{1}{3}e^{3x}\Big]_0^{\frac{1}{3}\ln 3}$$

$$\left(3\left(\tfrac{1}{3}\ln 3\right) - \tfrac{1}{3}e^{\ln 3}\right) - \left(0 - \tfrac{1}{3}e^{0}\right)$$

$$\ln 3 - \tfrac{1}{3}(3) + \tfrac{1}{3}$$

$$\ln 3 - 2/3$$

$$e^{3x} = 3$$

$$\ln 3 = 3x$$

$$\tfrac{1}{3}\ln 3 = x$$

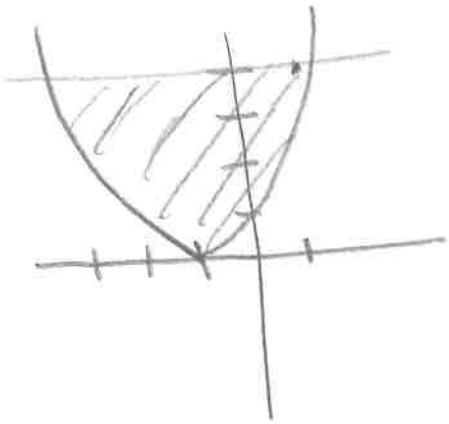
d) The region bounded by $y = (x+1)^2$ *and* $y = 4$ Sketch the graph

$$(x+1)^2 = 4$$

$$x+1 = \pm 2$$

$$x = -1 \pm 2$$

$$x = -3, 1$$



$$\int_{-3}^{1} \left(4 - (x+1)^2\right)dx$$

$$4x - \tfrac{1}{3}(x+1)^3\Big]_{-3}^{1}$$

$$\left(4 - \tfrac{1}{3}(2)^3\right) - \left(-12 - \tfrac{1}{3}(-2)^3\right)$$

$$4 - \tfrac{8}{3} + 12 - \tfrac{8}{3} = 16 - 16/3$$

$$16(1 - 1/3) = 16(2/3)$$

MATH 152 **QUIZ 1A** NAME Solutions

Mrs. Bonny Tighe 6.1

25 points TA_____ Wed 9/5/2018

1. Find the area of the region enclosed by the given curves. Sketch the graphs.

a) $y = \sin 2x$, *and* $y = \sin x$ on the interval $[0, \pi]$

$\sin 2x = \sin x$

$2\sin x\cos x - \sin x = 0$

$\sin x(2\cos x - 1) = 0$

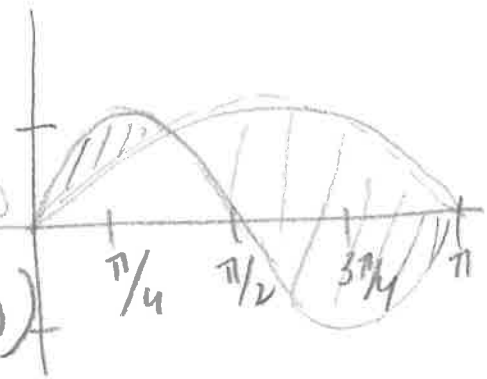
$\sin x = 0 \quad \cos x = 1/2$

$x = \pi/3$

$$\int_0^{\pi/3} (\sin 2x - \sin x)\,dx + \int_{\pi/3}^{\pi} (\sin x - \sin 2x)\,dx$$

$$-\tfrac{1}{2}\cos 2x + \cos x \Big]_0^{\pi/3} + -\cos x + \tfrac{1}{2}\cos 2x \Big]_{\pi/3}^{\pi}$$

$$\left(-\tfrac{1}{2}\cos 2\pi + \cos\pi\right) - \left(-\tfrac{1}{2}\cos\tfrac{2\pi}{3} + \cos\pi/3\right) + \left(-\cos\pi + \tfrac{1}{2}\cos 2\pi\right) - \left(-\cos\pi/3 + \tfrac{1}{2}\cos 2\pi/3\right)$$




$$\left(-\tfrac{1}{2} - 1\right) - \left(-\tfrac{1}{2}\left(-\tfrac{1}{2}\right) + \tfrac{1}{2}\right) + \left(1 + \tfrac{1}{2}\right) - \left(-\tfrac{1}{2} + \tfrac{1}{2}\left(-\tfrac{1}{2}\right)\right)$$

$$-\tfrac{3}{2} + \tfrac{1}{4} - \tfrac{1}{2} + \tfrac{3}{2} + \tfrac{1}{2} + \tfrac{1}{4} = 1/2$$

b) The region bounded by $x = y^2 - 4y$ *and* $x = 2y - y^2$

$y^2 - 4y = 2y - y^2$

$2y^2 - 6y = 0$

$2y(y - 3) = 0$

$y = 0, 3$

$$\int_0^3 (2y - y^2) - (y^2 - 4y)\,dy$$

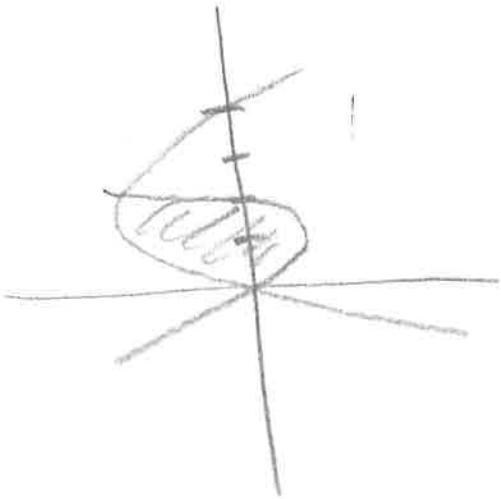
$$\int_0^3 (6y - 2y^2)\,dy =$$

$$3y^2 - \tfrac{2}{3}y^3 \Big]_0^3$$

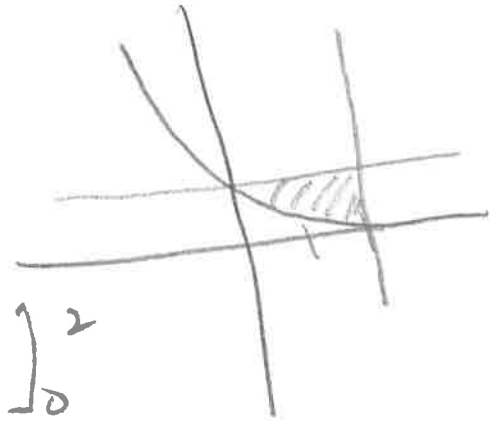
$$\left(3(9) - \tfrac{2}{3}(27)\right) - 0$$

$$27 - 18$$

$$9$$



c) $y = e^{-2x}$, $y = 1$ and $x = 2$ Sketch the graph



$e^{-2x} = 1$

$x = 0$

$$\int_0^2 (1 - e^{-2x})\,dx = x + \frac{1}{2}e^{-2x}\Big]_0^2$$

$$\left(2 + \frac{1}{2}e^{-4}\right) - \left(0 + \frac{1}{2}e^{-0}\right) = 2 + \frac{1}{2}e^{-4} - \frac{1}{2}$$

or

$$\frac{3}{2} + \frac{1}{2e^4}$$

d) The region bounded by $y = (x-2)^2$ *and* $y = 4$ Sketch the graph

$(x-2)^2 = 4$

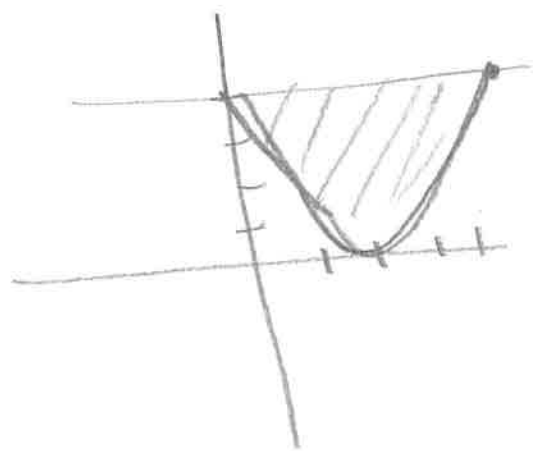
$x - 2 = \pm 2$

$x = 2 \pm 2$

$x = 4, 0$

$$\int_0^4 4 - (x-2)^2\,dx$$

$$4x - \frac{1}{3}(x-2)^2\Big]_0^4$$

$$\left(16 - \frac{1}{3}(2)^2\right) - \left(0 - \frac{1}{3}(-2)^2\right)$$

$$16 - \frac{4}{3} + \frac{4}{3} = 16$$